

Miter Bend Calculation Report

ASME B31.3-2018 | Internal pressure

DESIGN EQUATION

ASME B31.3-2018, 304.2.3

$q = T - c$; $r2 = (D - T)/2$; $\theta = \text{angle}/(2 \text{ joints})$

(4a) $P_m = S E W q / r2 * q / [q + 0.643 \tan(\theta) \sqrt{r2 q}]$

(4b) $P_m = S E W q / r2 * (R1 - r2) / (R1 - 0.5 r2)$

Multiple: $\min(4a, 4b)$, $\theta \leq 22.5 \text{ deg}$

Single: (4a) if $\theta \leq 22.5 \text{ deg}$; otherwise (4c)

(4c): use 1.25 instead of 0.643 in (4a)

(5) $R1 \geq A / \tan(\theta) + D/2$ (empirical A, not allowance c)

Pressure equations: P and S in MPa; dimensions in mm; input bar x 0.1 = MPa.

Project	QA miter
Prepared by	Device preview
Design reference	Illustrative coefficients only
Report date	2026-10-06 10:57

01 INPUTS

Design gauge pressure	20 bar
Miter cut angle θ	22.5 deg
Material / product	A106 Gr.B - user stress
Design metal temperature	20 C
S at design temperature	100 MPa; user supplied, A-1/A-1M
E / W	1 / 1
OD D / segment wall T	114.3 / 6.02 mm
Total allowances c	1 mm
Miter weld joint count	2
Effective radius R1	180 mm

02 RESULTS

Governing equation	Eq. 4a & 4b (multiple miter)
Maximum pressure P_m	49.4608 bar
Pressure margin $P_m - P$	29.4608 bar
Required R1 / wall extension M	117.5053 / 52.1329 mm
Minimum T meeting pressure	3.4978 mm; numerical solution

MEETS the implemented internal-pressure equations.

Provide the required extension M in the actual fabrication geometry; the app does not measure it. Overall design, material suitability, fabrication and other loads require independent verification.

04 REFERENCES AND LIMITS

Sources: B31.3-2018 304.2.3, supplied printed pp.22-23; B31.1-2022 104.2.3, supplied printed p.29. Inputs are user supplied. No external-pressure, fatigue, remaining-life or complete code qualification is performed.